

EqualityConstraint#

https://pytorch.org/docs/stable/generated/torch.fx.experimental.symbolic_shapes

Description:

Represent and decide various kinds of equality constraints between input sources. A “source pair” is a pair of input sources for dynamic dimensions that are specified equal. We represent source_pairs in a union-find forest so that we can efficiently check whether two such sources are transitively equal. A “derived equality” relates an input source to an expression over a root. The root can be another input source, corresponding to some dynamic dimension, or a phantom symbol that does not directly represent any dynamic dimension. We represent derived_equalities involving input sources in a transitively-closed map so that we can efficiently check whether an input source is transitively equal to a given expression over another input source. (NOTE: In contrast, it is easy to decide whether an input source is transitively equal to a given expression over a phantom symbol; such expressions are already in canonical form and so the problem reduces to symbolic expression equality.)

Function Signature

```
class
torch.fx.experimental.symbolic_shapes.EqualityConstraint(war
n_only, source_pairs, derived_equalities, phantom_symbols,
relaxed_sources)[source]#
```

Detailed Documentation

Documentation

Represent and decide various kinds of equality constraints between input sources.

A “source pair” is a pair of input sources for dynamic dimensions that are specified equal. We represent `source_pairs` in a union-find forest so that we can efficiently check whether two such sources are transitively equal.

A “derived equality” relates an input source to an expression over a root. The root can be another input source, corresponding to some dynamic dimension, or a phantom symbol that does not directly represent any dynamic dimension. We represent `derived_equalities` involving input sources in a transitively-closed map so that we can efficiently check whether an input source is transitively equal to a given expression over another input source. (NOTE: In contrast, it is easy to decide whether an input source is transitively equal to a given expression over a phantom symbol; such expressions are already in canonical form and so the problem reduces to symbolic expression equality.)

